Energy Consumption Forecasting

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DSAI 302 Capstone Project | MEF University

Project Overview

Objective

Stability & Reliability: Precise matching of generation with demand is vital for power grid stability.

Key Results

Multivariate LSTM emerged as the definitive leader (R^2=0.996), outperforming **LightGBM** (R^2=0.927) and traditional statistical models.

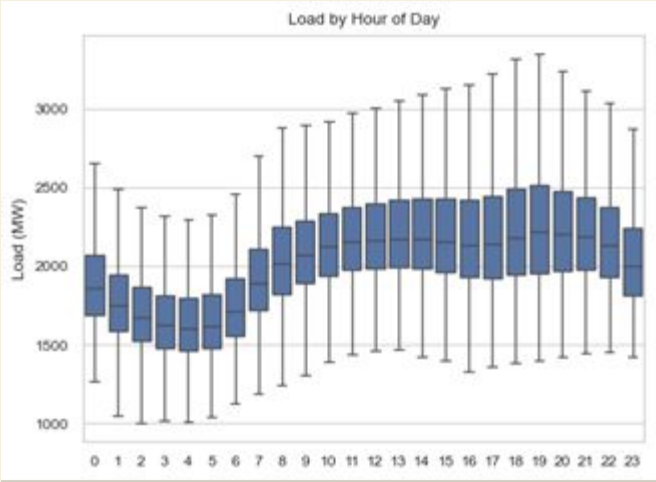
Challenges

Non-linear behaviors driven by climate.
Highly autoregressive temporal dynamics.
Multi-layered seasonal patterns.

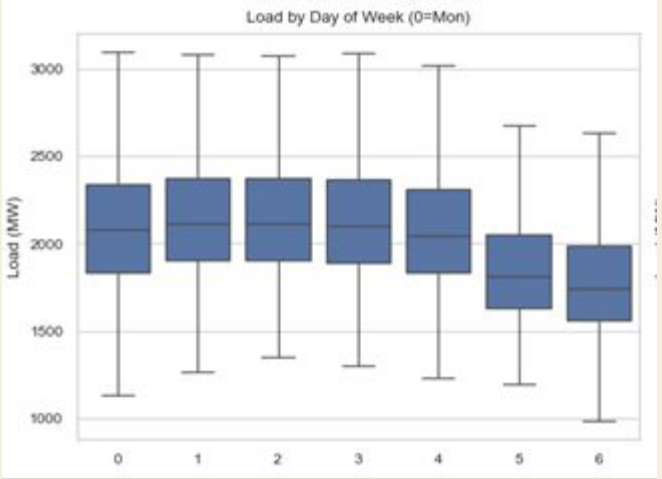
Data & Preprocessing

Metric	Dataset Value
Time Period	October 2004 – August 2018
Observations	121,271 Hourly Records
Average Load	2,037.87 MW
Peak Demand	3,746.00 MW
Stationarity	Observed Long-term Stationarity

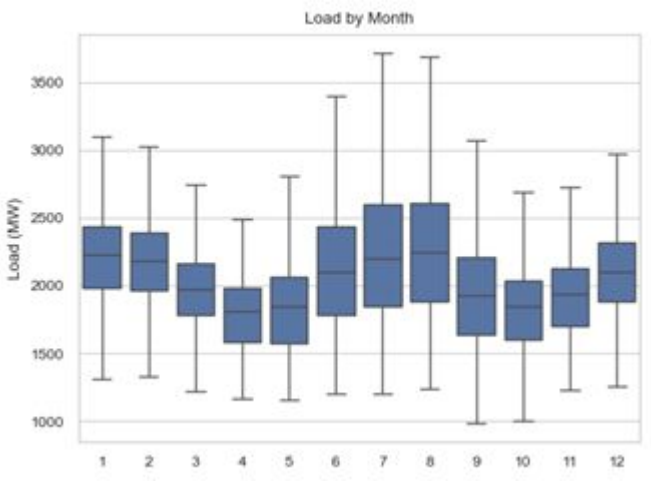
Seasonality Profiles (EDA)



Intra-day (Hourly)



Business Cycle (Weekly)



Annual (Monthly)

Preprocessing Pipeline

- **Chronological Sorting:** Maintains sequential integrity to avoid data leakage.
- **Cyclical Encoding:** sin/cos transformations for hour and day features.
- **Autoregressive Lags:** Lags (24h, 168h) capture historical momentum.
- **Rolling Stats:** 24h moving average with strict look-ahead prevention.
- **Normalization:** MinMaxScaler to stabilize gradient updates.

Methods

Tree-Based Ensembles

RF, XGBoost, LightGBM: Optimized via TimeSeriesSplit CV. Heavy reliance on engineered autoregressive lag variables.

Prophet

Additive regression model handling trend and seasonality. Integrated US Holidays as exogenous structural breaks.

LSTM

Designed to process multivariate tensors over a 24-hour receptive field. Implicitly learns dependencies through backpropagation through time.

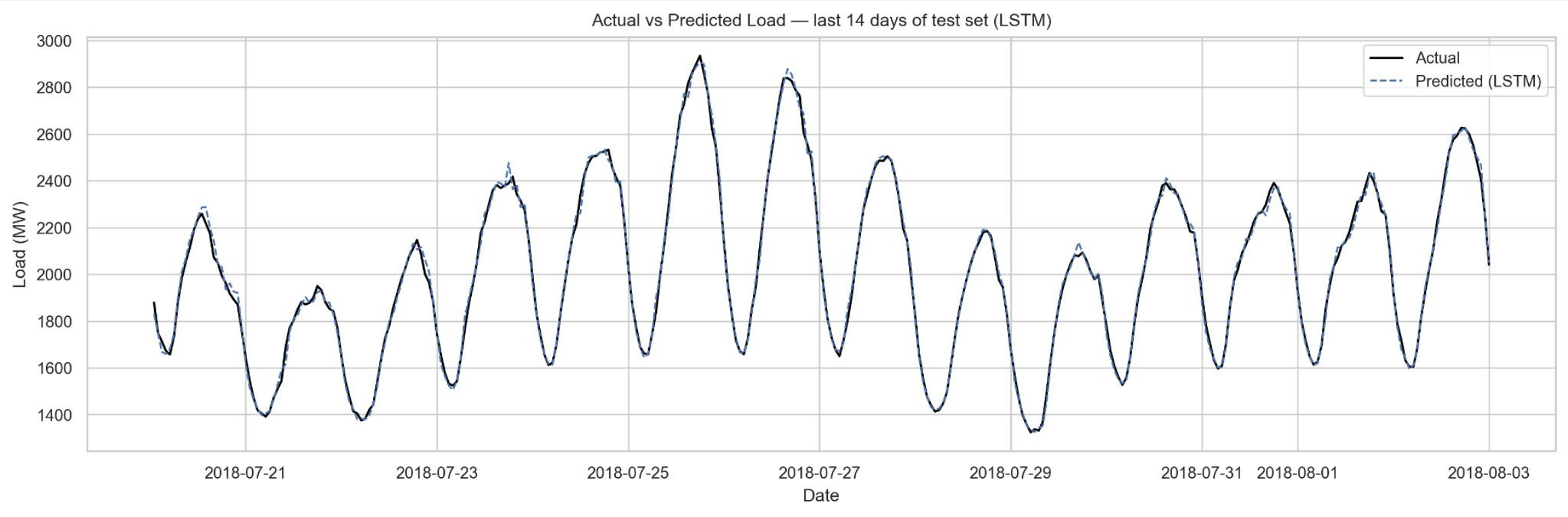
Results

Performance Benchmark

Model	MAE	MAPE (%)	RMSE	R^2
LSTM	**16.84**	**0.83**	**24.63**	**0.996**
LightGBM	78.09	3.80	104.23	0.927
XGBoost	79.64	3.87	106.66	0.924
Prophet	221.21	11.26	272.84	0.500

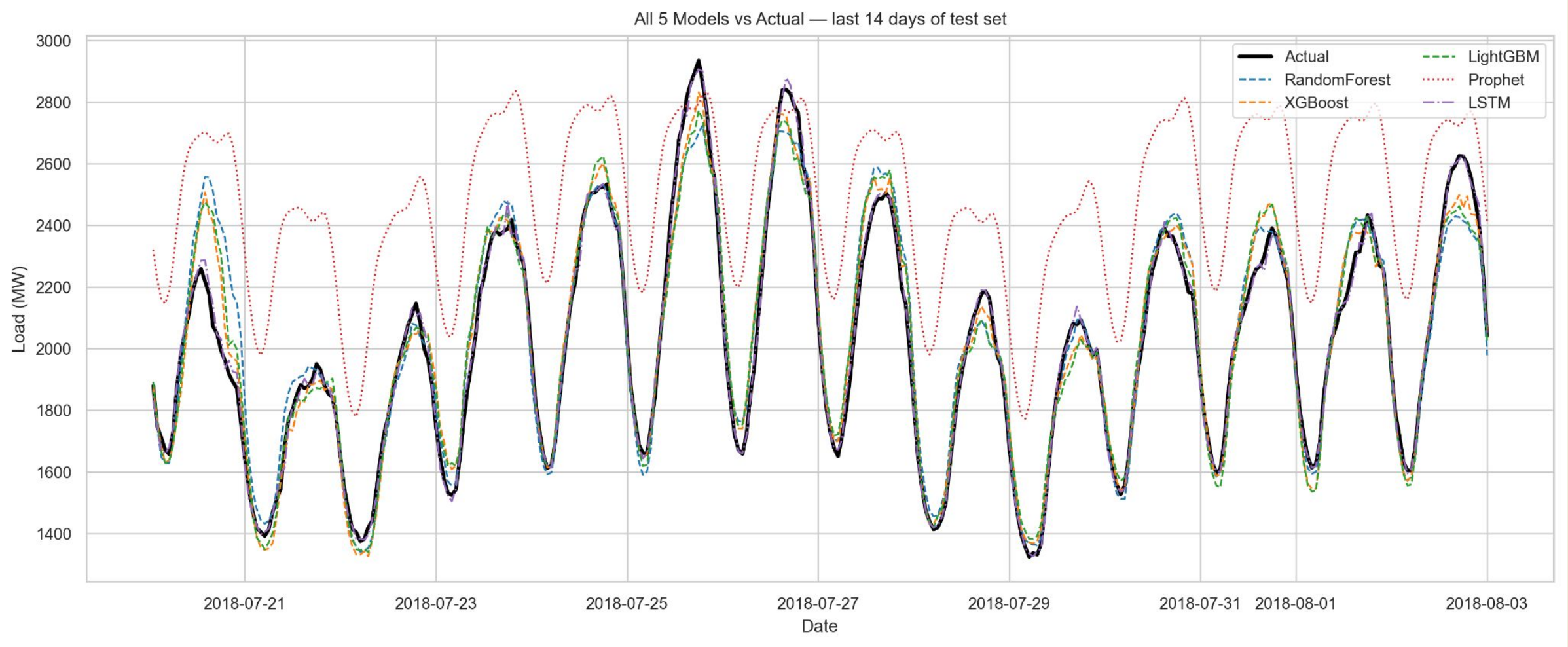
** LSTM achieved an average forecast error of less than 1% relative to actual load.*

Visual Accuracy (Actual vs Predicted)

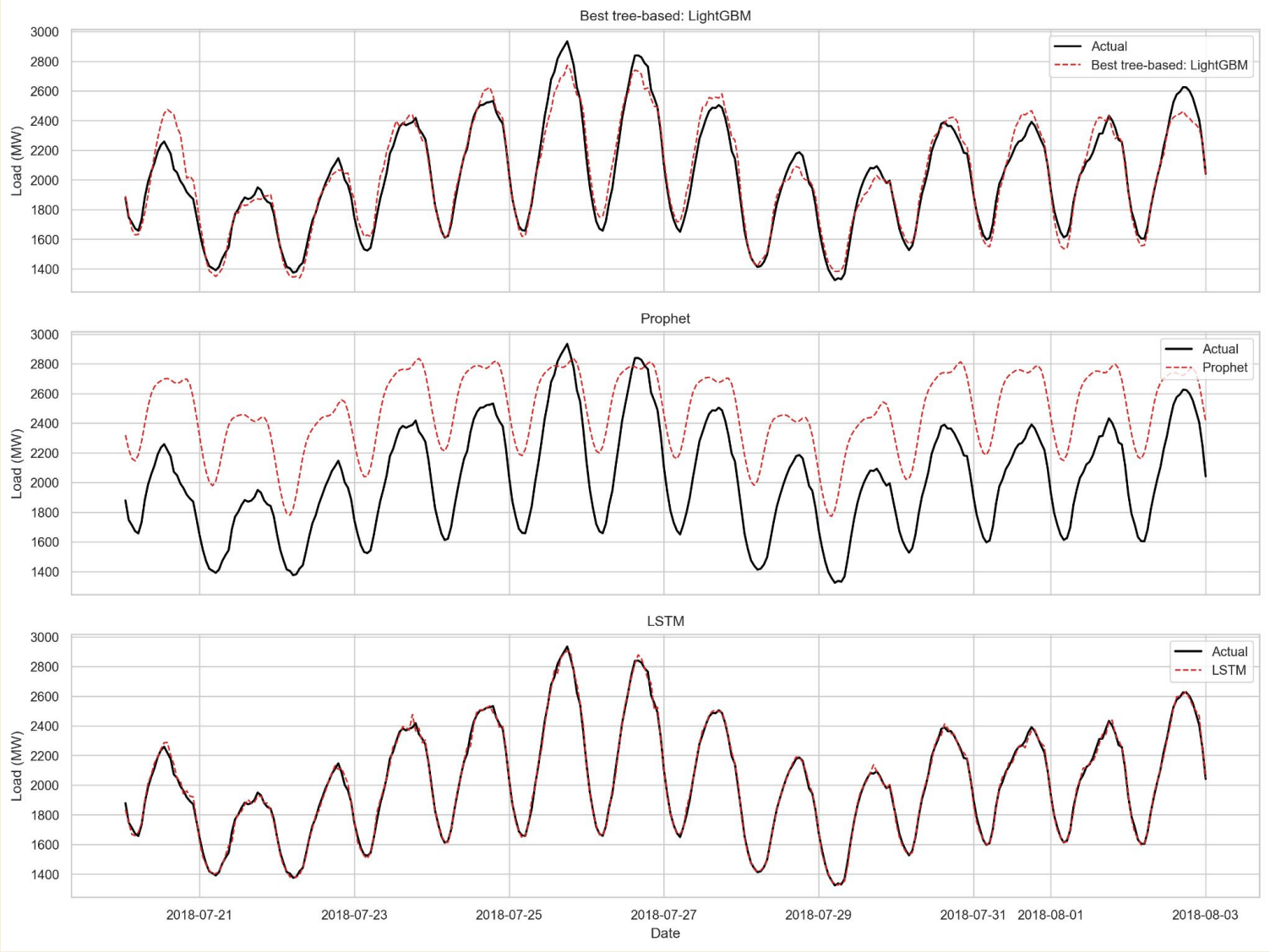


LSTM's gated memory tracks high-frequency diurnal peaks and nocturnal troughs with near-perfect overlap.

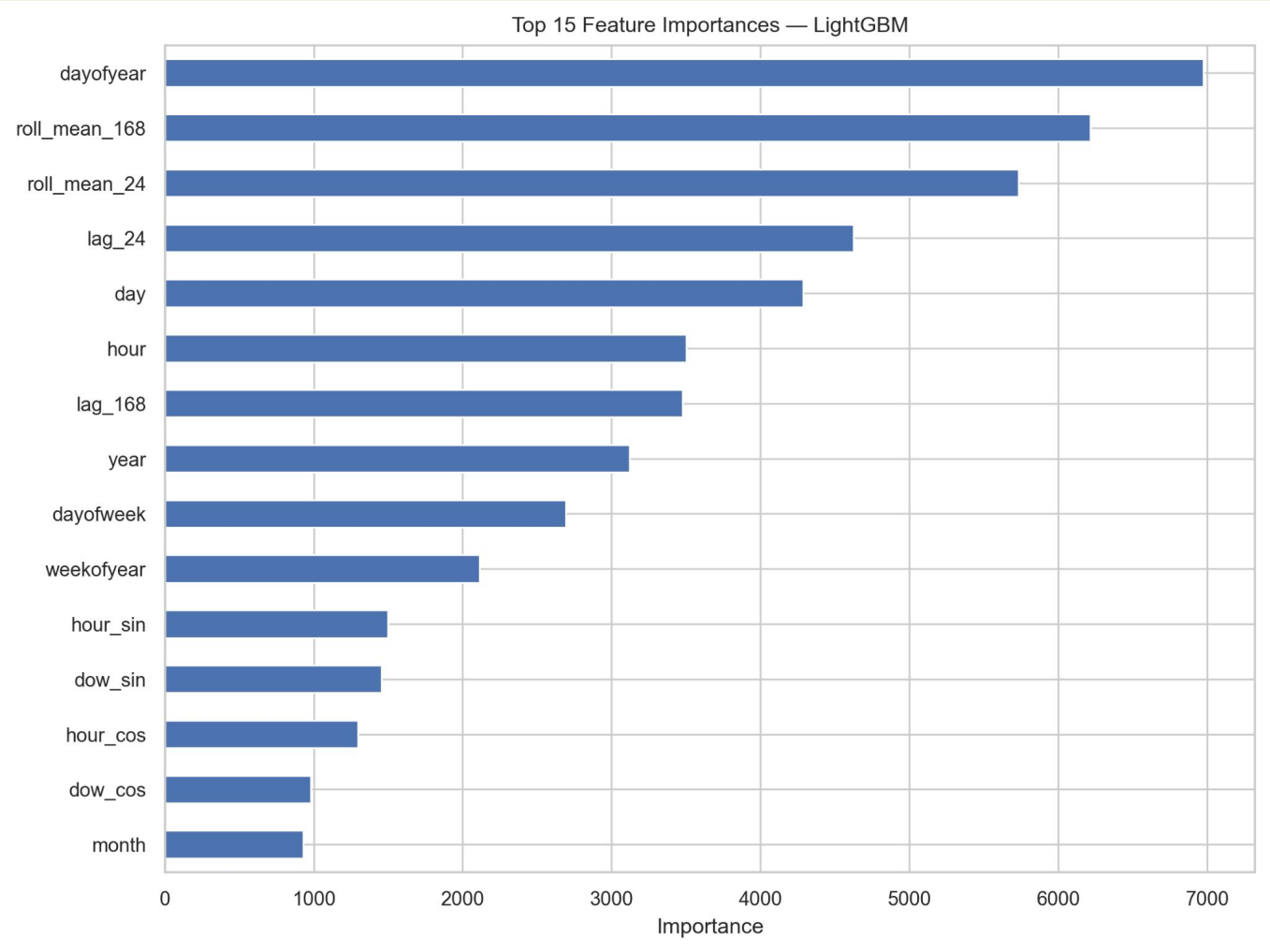
Model Comparison



Paradigm Comparison



Feature Importance



Keys to Forecasting Success

Sequential Memory

LSTMs internalize the manifold of demand patterns via hidden states rather than static features.

Feature Integrity

Tree models require meticulous manual engineering (lags/rolling stats) to compensate for lack of sequential memory.

Grid Impact

A sub-1% MAPE enables operators to reduce waste and optimize regional dispatch infrastructure.

Thanks!

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